

Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

COURSE AT A GLANCE

Review Content

REVIEW

3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 5.2 Properties of Exponents and Scientific Notation

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation and write an equation from given information.
- I can apply a linear function to a context.
- I can identify the solution of a linear system as an intersection point.
- I can solve a simple two-variable linear system using an appropriate method.
- I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

3.5 Relations and Functions; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

- I can evaluate a function from a formula, graph, or table.
- I can determine whether a relation is a function.
- I can determine domain and range from a graph or table.
- I can solve for an input that produces a specified output.
- I can interpret inputs, outputs, domain, and range in context.
- I can calculate and interpret average rate of change.
- I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

UNIT LEARNING OBJECTIVES

- I can add and subtract polynomials.
- I can multiply polynomials.
- I can factor out a greatest common factor.
- I can factor a quadratic trinomial.
- I can factor a difference of squares and other required special cases.
- I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

UNIT LEARNING OBJECTIVES

- I can state excluded values for a rational expression.
- I can simplify a rational expression while preserving its restrictions.
- I can multiply and divide rational expressions.
- I can add and subtract rational expressions using a common denominator.
- I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

8.3 Simplify Rational Exponents; 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

UNIT LEARNING OBJECTIVES

- I can simplify expressions containing negative exponents.
- I can rewrite expressions between radical and rational-exponent forms.
- I can simplify radical expressions.
- I can apply exponent rules to expressions with integer and rational exponents.
- I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations

UNIT LEARNING OBJECTIVES

- I can identify key features of a quadratic function from its graph or equation.
- I can connect standard, factored, and vertex forms to features of a parabola.
- I can solve a quadratic equation by factoring.
- I can solve a quadratic equation using the square-root property.
- I can solve a quadratic equation by completing the square.
- I can solve a quadratic equation using the quadratic formula.
- I can choose an efficient method for solving a quadratic equation.
- I can graph a quadratic function using its vertex, intercepts, and symmetry.
- I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

9.3 Solve Quadratic Equations Using the Quadratic Formula

UNIT LEARNING OBJECTIVES

- I can perform basic arithmetic with complex numbers.
- I can express the nonreal solutions of a quadratic equation using complex numbers.